Philipps-Universität Marburg
Fachbereich Mathematik & Informatik

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Probabilistic Combinatorics SS 24

Sheet 1

Due: 22 April, 2024 at 19:00

(1) Let $R(\ell, k)$ be the Ramsey number as it was defined in the lecture.

(i) Show that if $\ell, k \geqslant 2$, then $R(\ell, k) \leqslant R(\ell - 1, k) + R(\ell, k - 1)$.

(ii) Conclude Ramsey's theorem (Theorem 2.2 in the lecture notes) with (i).

(2) Let $n \geqslant 4$ and $\mathcal{A}$ be a set of n-element subsets of $\mathbb{N}$ satisfying $\#\mathcal{A} \leqslant \frac{4^{n-1}}{3^n}$. Show via a probabilistic argument: There is a map $f : \mathbb{N} \to [4]$ such that $A \in \mathcal{A}$ $\#f(A) = 4$.

(3) Let $n \geqslant 2$ and $\mathcal{A}$ be a set of n-element subsets of $\mathbb{N}$ with $\#\mathcal{A} < 2^{n-1}$. Show: There is a map $f : \mathbb{N} \to [2]$ such that there is no set $A \in \mathcal{A}$ with $\#f(A) = 1$.

(4) Consider an 100×100 chessboard such that each square is labeled by an integer in $[5000]$ and each number appears on exactly two squares.

Prove that is possible to choose 100 squares such that the numbers on the 100 squares are different and no two of the selected square are in the same column or the same row.